

A new approach to locate the voltage sag source using real current component

Noraliza Hamzah*, Azah Mohamed, Aini Hussain

*Faculty of Engineering, Department of Electrical Electronic and System Engineering,
Universiti Kebangsaan Malaysia, Bangi, 43600 Selangor, Malaysia*

Received 23 November 2003; received in revised form 17 March 2004; accepted 18 March 2004
Available online 10 June 2004

Abstract

Voltage sag can cause hours of downtime, substantial loss of product and also can attribute to malfunctions, instabilities and shorter lifetime of the load. Accurate voltage sag source location can help to minimize the loss and problems caused by voltage sag in a power distribution system. This paper proposes a new method to locate the source of voltage sag in a power distribution system. The proposed method uses the polarity of the real current component to determine the sag location relative to the monitoring point. The product of the RMS current and the power factor angle at the monitoring point is employed for the sag source location. A graph of this product against time is plotted. The voltage sag source location is determined by examining the polarity of the RMS current at the beginning of the sag. The proposed method has been verified by simulations and the results are proven to be in agreement when compared with the slope of system trajectory method.

© 2004 Elsevier B.V. All rights reserved.

Keywords: Voltage sag; Source location; Distribution system; Real current

1. Introduction

Voltage sag is a temporary decrease in the RMS voltage magnitude between 0.1 and 0.9 p.u. and with duration of mostly less than 1 s. Its frequency of occurrence is between a few tens and several hundreds times per year [1]. It is the most important power quality problem facing many industrial customers since equipment used in modern industrial plants such as process controllers and adjustable speed drives is becoming more sensitive to voltage sag. The causes of voltage sags are fault conditions, motor starting, transformer energizing and other sudden load changes. Voltage sags are typically caused by fault conditions [2], in which short-circuit faults and earth faults are found to cause severe voltage sags [3]. In industrial and commercial power systems, faults on one feeder tends to cause voltage drops on all other feeders in the plant [4]. During short-circuit faults, voltage sags occur whenever fault current flows through a fault impedance. Voltage returns to normal as soon as a fault-clearing device interrupts the flow of current. These faults may be far from the interrupted process, but close

enough to cause problems throughout the system. Even when voltage returns to normal, many sensitive loads experience a production outage if the voltage sag magnitude and duration are outside the load ride-through capabilities.

Locating the source of voltage sag is important before any voltage sag mitigation technique is done to eliminate the sag. A wrong mitigation solution can aggravate the voltage sag problem because only after information about a voltage sag source location is available, can power-quality trouble-shooting, diagnosis and mitigation be carried out. The advantage of locating the source of voltage sag is that any disputes among the major responsibility party can be resolved fairly [5].

To date only two references cite the methods to locate the sources of voltage sags from the literature. A method using the disturbance power and disturbance energy to determine which side of a recording device the voltage sag originates is based on the concept that active power tends to flow away from a nonlinear load [6]. This concept is translated in terms of disturbance power and disturbance energy to determine on which side of a recording device the voltage sags originate. The directions of the disturbance energy as well as the disturbance of real power flow are used to locate the voltage sag source. The method will rely on the degree of

* Corresponding author. Tel.: +603-8921-6322; fax: +603-8921-6146.
E-mail address: noraliza@vlsi.eng.ukm.my (N. Hamzah).

confidence of both the disturbance power and disturbance energy. Thus, the degree of confidence will be reduced if results from disturbance energy and disturbance power do not match. Another most recent technique to locate the origin of voltage sag is by employing the slope of the line-fitting parameters of current and voltage during voltage sag [5]. The method plots the product of voltage magnitude and power factor against current magnitude at a particular measurement point. A line-fitting of the measured points are performed and the sign of the slope indicates the direction of voltage sag source. A positive slope shows that the sag is from upstream and a negative slope shows that it is from downstream. The method has only been verified using three-phase-to-ground faults.

This paper proposes a new method which is based on the directional relay concept [7] where the phase angle difference between current and voltage or power factor angle is used to determine the voltage sag source. In the method, magnitude of currents and phase angles of voltages and currents are measured at the measuring point. The current magnitude is then multiplied with the cosine of the power factor angle and the product is then plotted against time. The product polarity is used to indicate the direction of voltage sag source either it is from upstream or downstream relative to the measuring point. A positive polarity of the product at the beginning of the sag duration will indicate the voltage sag source is from downstream. On the other hand, a negative polarity of the product at the beginning of the sag duration will indicate that the voltage sag source is from upstream.

The proposed method is verified on a test distribution system modeled using an electromagnetic transient program EMTDC/PSCAD and the data are processed via MATLAB codes.

2. Derivation of voltage sag source location method

The basis of the proposed voltage sag source location considers fault analysis at pre-fault and during fault conditions. First, consider a two-source system shown Fig. 1.

From Fig. 1, the current flow from E_1 to E_2 at a pre-fault condition is given by

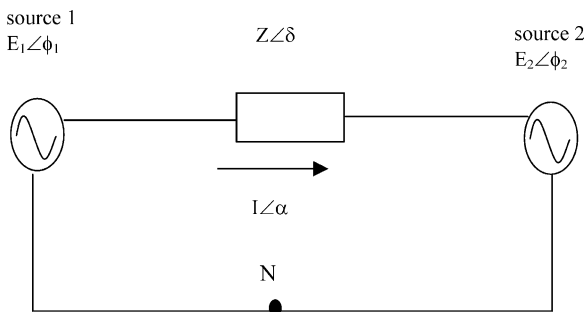


Fig. 1. A two-source system at pre-fault condition.

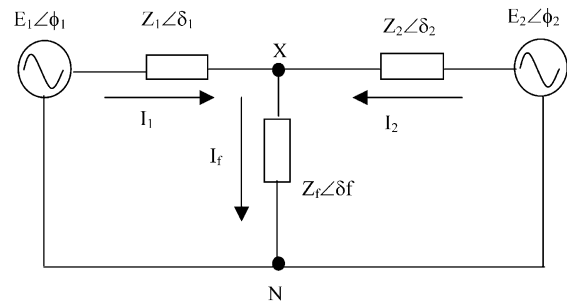


Fig. 2. A two-source system at a fault condition.

$$I\angle\alpha = \frac{E_1\angle\phi_1 - E_2\angle\phi_2}{Z\angle\delta} \quad (1)$$

where $Z\angle\delta$ is the line impedance, $E_1\angle\phi_1$ and $E_2\angle\phi_2$ are the voltage sources for sources 1 and 2, respectively. It is noted that before a fault there is only one loop and the only current that flows is $I\angle\alpha$ if $E_1 > E_2$. If a fault occurs at X with a fault impedance of $Z_f\angle\delta_f$ as shown in Fig. 2, two current loops will be generated. A fault at X will cause a voltage measured at X to be very small which is almost zero [3], and currents I_1 , I_2 and I_f will be generated, where I_1 is the current from the source $E_1\angle\phi_1$, I_2 is the current from the source $E_2\angle\phi_2$ and I_f is the current flowing through the impedance of the short-circuit path to ground. In terms of current directions, I_1 is in the same as the direction as the pre-fault current $I\angle\alpha$ since it is from the same source E_1 and flows towards X. If impedance Z_2 is much greater than the fault impedance Z_f , then current $I_2 \approx 0$ and current from the source E_1 will flow to Z_f . Otherwise, I_2 is seen as the current coming from E_2 . The concept of these current directions as to whether it is seen coming from E_1 or E_2 during a fault will be applied in the development of an indicator for locating the source of voltage sag.

2.1. The real current component concept

To illustrate further the proposed voltage sag source location method, an equivalent circuit representing the two-source system is considered as shown in Fig. 3. Consider a fault at X and two monitoring points M_A and M_B .

For a fault at X, using Kirchoff's second law, the voltage at M_A will be

$$\tilde{V} = \tilde{E}_1 - \tilde{I}\tilde{Z}_1 \quad (2)$$

where $\tilde{V}$ and $\tilde{I}$ are the voltage and current measured directly at M_A , respectively. Multiply on both sides of Eq. (2) by $\tilde{I}^*$

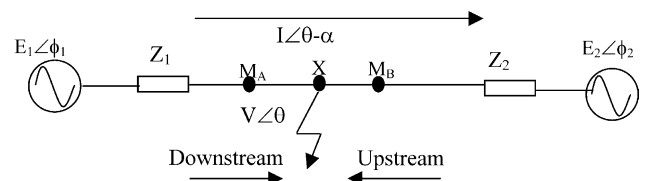


Fig. 3. A two-source system for the voltage sag source location analysis.

will lead to

$$\tilde{V}\tilde{I}^* = \tilde{E}_1\tilde{I}^* - I^2\tilde{Z}_1 \quad (3)$$

Extracting the real part of Eq. (3) will lead to

$$VI\cos(\theta - \alpha) = E_1I\cos(\phi_1 - \alpha) - I^2R \quad (4)$$

where θ and α are the phase angles for voltage and current at the measuring point M_A , respectively. The phase difference $(\theta - \alpha)$ is the power factor angle at the monitoring point.

From (4), for the case of monitoring at a point before the fault X, i.e. M_A , the current that flows from E_1 to X, $I\cos(\theta - \alpha) > 0$. The fault is said to occur at downstream location as seen from the point M_A and the voltage sag source is seen coming from downstream. This current is in the same direction as the pre-fault current from E_1 to E_2 . In the case of monitoring at point after the fault X, i.e. M_B , the current $I\cos(\theta - \alpha) < 0$ because the current already flows to ground at the fault point and this current is seen as current flowing from E_2 to X. The fault is said to occur at upstream location as seen from the monitoring point M_B and the voltage sag source is seen coming from upstream.

In the proposed method the value of fault impedance is taken into consideration because if a fault impedance, $Z_f \ll Z_2$, then no current is seen flowing from E_2 since all current flows through the fault path, from the source E_1 to Z_f . However, at the beginning of a voltage sag, the current can be significantly higher than the steady-state current due to a sudden change in electrical condition [8]. Hence, even if a fault impedance is very small at the beginning of a fault a sudden change in current direction can still be observed. Therefore a more suitable indicator chosen for voltage sag source location will be based on the direction of current at the beginning of a fault. The condition for the proposed method is that if at the beginning of a fault, $I\cos(\theta - \alpha) > 0$, the source of voltage sag is from downstream. On the other hand, if $I\cos(\theta - \alpha) < 0$, the source of voltage sag is from upstream. The concept of the direction of real current components used for voltage sag source location can also be applied to a single source system such as a radial distribution system.

3. Implementation of the proposed method

The proposed method to locate the source of voltage sags is verified on a test distribution system modeled using an electromagnetic transient program PSCAD/EMTDC. The procedure implemented is as follows:

- (i) Create a voltage sag condition by simulating a short-circuit fault at a specified location.
- (ii) Obtain the magnitude and phase of voltage and current from the measuring meter at pre-fault and during fault times.
- (iii) Calculate the values of $I\cos(\theta - \alpha)$ for a few cycles of pre-fault and during fault durations.

- (iv) Graphically plot coordinates of $I\cos(\theta - \alpha)$ against time of a few cycles of pre-fault and during fault durations.
- (v) Check the polarity of $I\cos(\theta - \alpha)$ at the beginning of fault. If it is positive, the source of voltage sag is from downstream. On the other hand, if it is negative, the source of voltage sag is from upstream.

The process from (ii) to (iv) is done via MATLAB programming codes.

There are two sets of test system used to verify the proposed method in this paper. The first one is the test distribution system is as shown in Fig. 4(a) in which it is fed from a 33 kV, 15 MVA source at 50 Hz frequency. The system has four main feeders with each feeder connected to several step-down transformers according to the load sizes. Three fault points are considered, namely F1, F2 and F3 and three monitoring points are marked as M1, M2 and M3. Two types of faults are considered in this simulation, namely balanced and unbalanced faults. Faults are simulated through a resistance to ground and the duration of the fault is 0.3 s. The two types of unbalanced faults simulated are single line to ground fault (SLG) and double line to ground (DLG) fault. The current, voltage and their phase angles are measured at M1, M2 and M3.

The second test system is as shown in Fig. 4(b). The first source is 33 kV, 15 MVA and the second source is 3 kV, 7.5 MVA. Two fault points are considered, namely FA1 and FB1 and four monitoring points are marked as MA1, MA2, MB1 and MB2. Two types of faults are considered in this simulation, namely balanced and unbalanced faults.

4. Results and comparison with the slope of system trajectory method

In this section simulation results of the proposed real current component method for voltage sag source location are presented for one-source and two-source system. Case A presents the results considering the balanced three-phase faults whilst case B presents results due to unbalanced faults. Results obtained using the system trajectory method [5] are also presented for comparison purposes.

4.1. Results from one-source system

For one-source system, balanced and unbalanced fault cases are considered.

4.1.1. Case A: balanced fault

The results of three-phase balanced faults created at F1, F2 and F3 are first presented and analyzed. Table 1 tabulates the faulted bus locations.

Fig. 5(a)–(c) shows the graph of real current components plotted against time for a fault at F1 measurements made at M1, M2 and M3, respectively. In Fig. 5(a), it can be seen that the real current settles out to a positive value

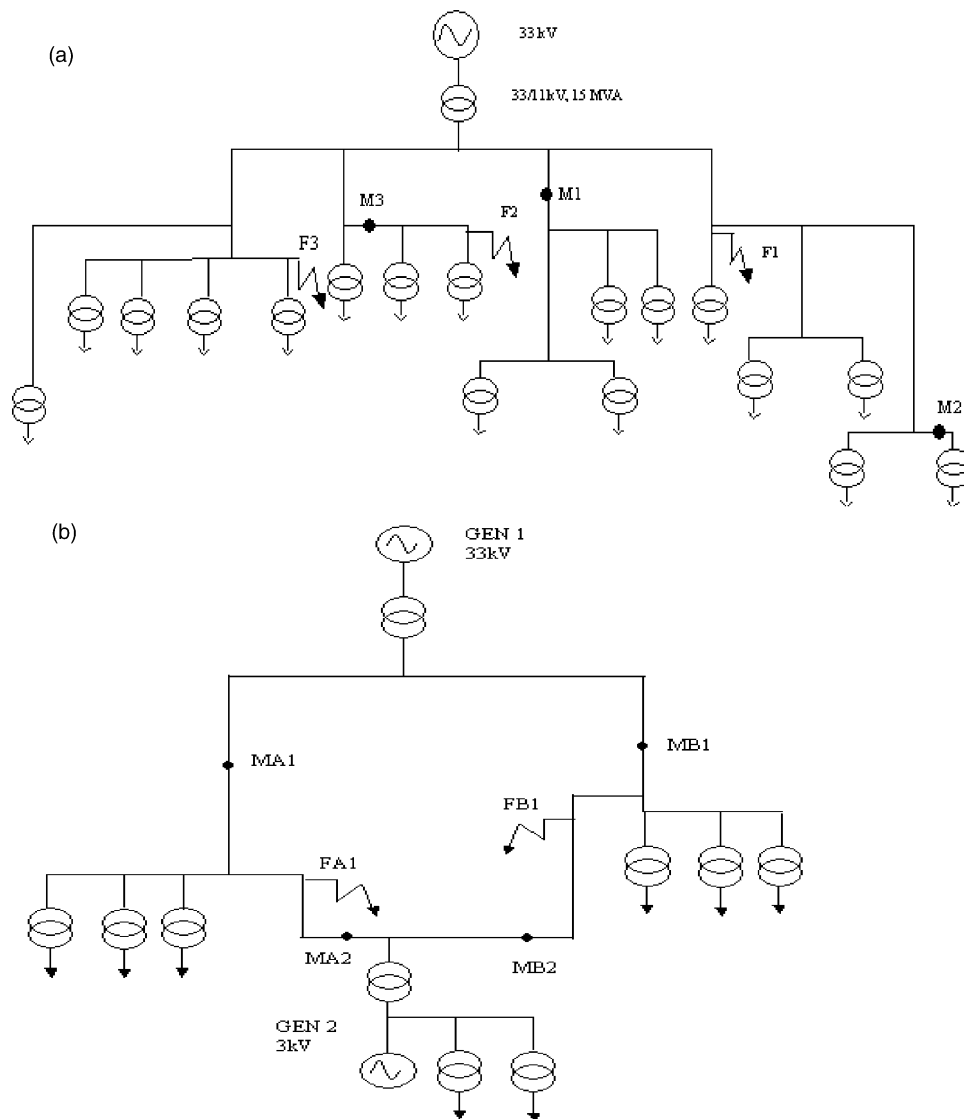


Fig. 4. Test system (a) for voltage sag source location and (b) for two-source voltage sag location.

from the beginning of the sag, indicating the voltage sag source is from downstream. Fig. 5(b) and (c) shows that the real current settles out to a negative value at the beginning of sag, indicating that the voltage sag source is from upstream. Fig. 6 shows the results from the method of sys-

tem trajectory slope [5]. In Fig. 6(a), a negative slope is noticed indicating the voltage sag source is from a downstream location whilst both graphs in Fig. 6(b) and (c) show a positive slope, indicating an upstream voltage sag source location.

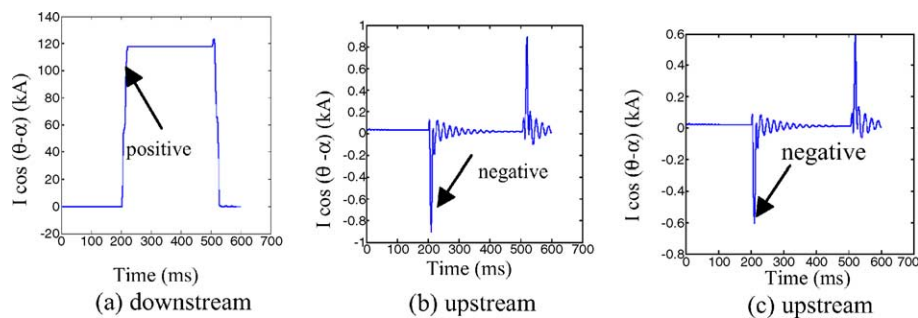


Fig. 5. Fault at F1 for monitoring at (a) M1, (b) M2 and (c) M3 using the real current component.

Table 1
Balanced fault bus details for one-source system

Fault bus	Monitoring buses	
	Downstream fault	Upstream fault
F1	M1	M2, M3
F2	M3	M1, M2
F3	–	M1, M2, M3

Fig. 7(a)–(c) shows the real current components for a fault at F2 and measurements made at M1, M2 and M3. In Fig. 7(a) and (b) the real current settles out to a negative value at the beginning of the sag, indicating the correct voltage sag source location which is from upstream. On the other hand, in Fig. 7(c), the real current settles out to a positive value at the beginning of the sag, indicating a downstream voltage sag source location at downstream. Meanwhile Fig. 8(a)–(c) shows the results from method of system trajectory slope [5]. In Fig. 8(a) and (b), a positive slope is noticed indicating

an upstream voltage sag source location whilst in Fig. 8(c) a negative slope is observed, indicating a downstream voltage sag source location.

It can be noted that for upstream faults a spike is seen at the beginning of the fault before the current settles to zero. In one-source system, the only voltage source available is E_1 . Hence, when fault occurs on the left-hand side of the measuring point (M_B), no voltage is available to push current from the negative side. Therefore at the beginning of voltage sag, a high reverse current direction is seen before it settles to zero which appears as a spike graph for upstream cases.

Fig. 9(a)–(c) shows the real current components for a fault at F3. In all these figures, the real current settles out to a negative value at the beginning of the sag indicating the correct upstream direction of voltage sag source. Fig. 10(a)–(c) shows the results from the method of system trajectory slope [5]. A positive slope noticed in Fig. 10(a)–(c) indicates the voltage sag source is from upstream location.

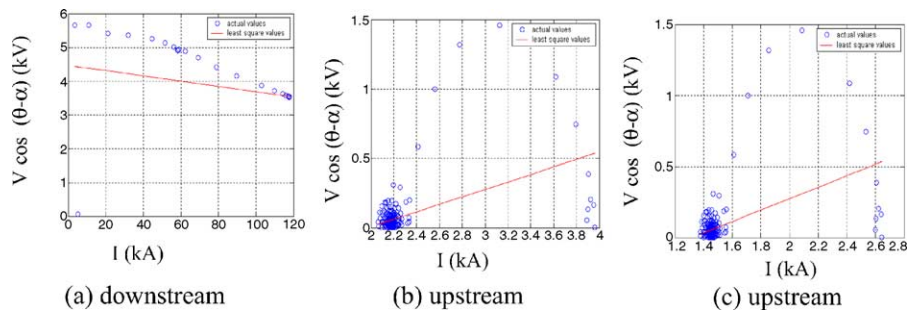


Fig. 6. Fault at F1 for monitoring at (a) M1, (b) M2 and (c) M3 using the slope method.

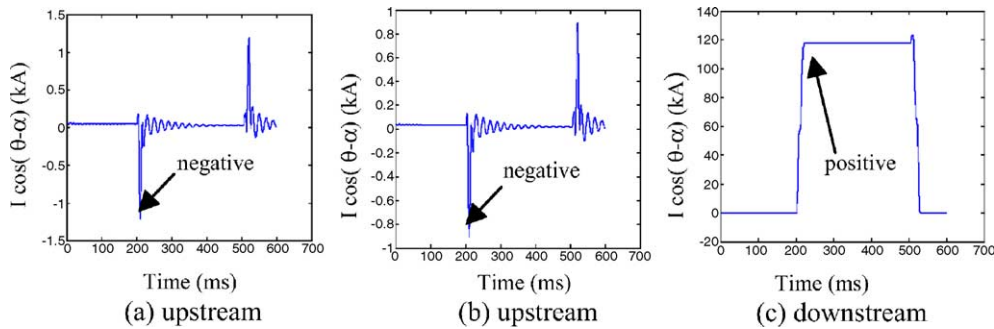


Fig. 7. Fault at F2 for monitoring at (a) M1, (b) M2 and (c) M3 using the real current component.

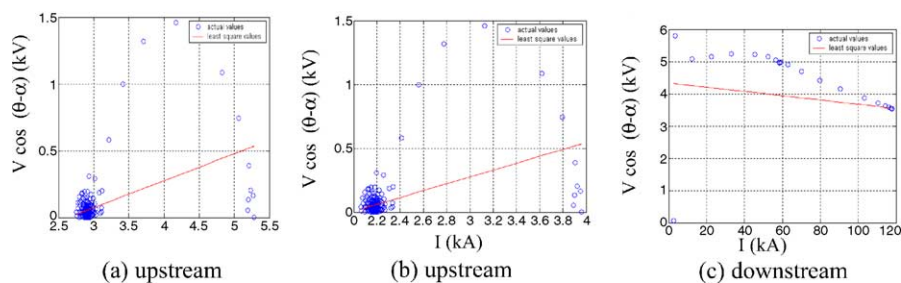


Fig. 8. Fault at F2 for monitoring at (a) M1, (b) M2 and (c) M3 using the slope method.

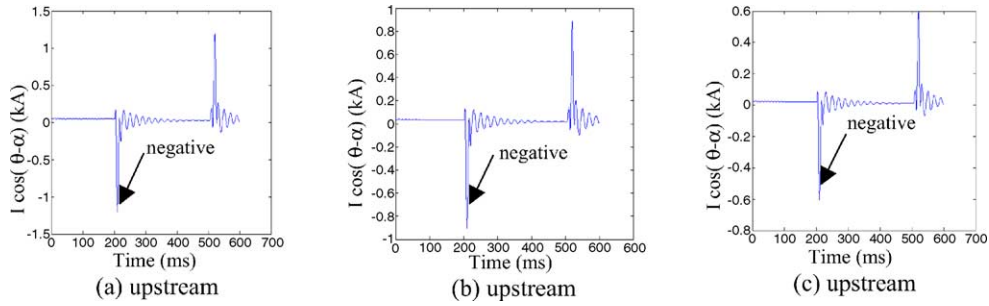


Fig. 9. Fault at F3 for monitoring at (a) M1, (b) M2 and (c) M3 using the real current component.

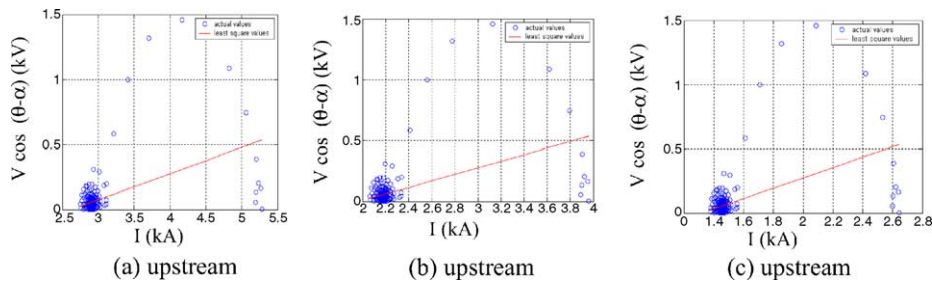


Fig. 10. Fault at F3 for monitoring at (a) M1, (b) M2 and (c) M3 using the slope method.

From Figs. 5(a) and 7(c) it is observed that in downstream events, the values of currents remain positive throughout the sag period. However, for upstream event, there is a significant increase in real current component at the beginning of the sag but at opposite polarity from the current at the pre-sag event. At the beginning of the voltage sag, the current decreases significantly and the steady-state currents settle out to almost zero value. This can happen when the value of fault impedance Z_f is very small where a significantly high currents flow through Z_f . The results from the slope of system trajectory method show that for upstream events, the actual data values do not distribute evenly along the line-fitting. More data values are seen concentrated at the beginning of the sag and the actual pattern is rather seen as a curve than a straight line. Comparing the results of both methods, it is proven that the results are in agreement in which both methods locate the correct voltage sag sources. However, the disadvantage of the slope of system trajectory methods is that it needs a line-fitting technique to correlate

Table 2

Unbalanced fault bus details for one-source system

Fault type	Monitoring buses	
	Downstream fault	Upstream fault
SLG	M3	M1, M2
DLG (phase A)	M3	M1, M2
DLG (phase B)	M3	M1, M2

the points of $(I, V \cos(\theta - \alpha))$ and the line-fitting can only give an approximation of the system trajectory slope.

4.1.2. Case B: unbalanced faults

In this section, results from the two types of unbalanced fault created at F2 are presented. Table 2 tabulates the faulted bus locations.

Fig. 11(a)–(c) shows the results for an SLG fault created at F2 and measurements made at M1, M2 and M3. In Fig. 11(a) and (b) both the real currents settle out to negative

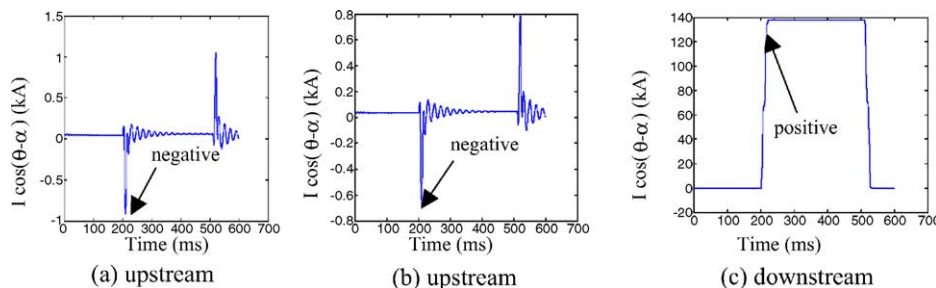


Fig. 11. Fault at F2 (SLG) for monitoring at (a) M1, (b) M2 and (c) M3 using the real current component.

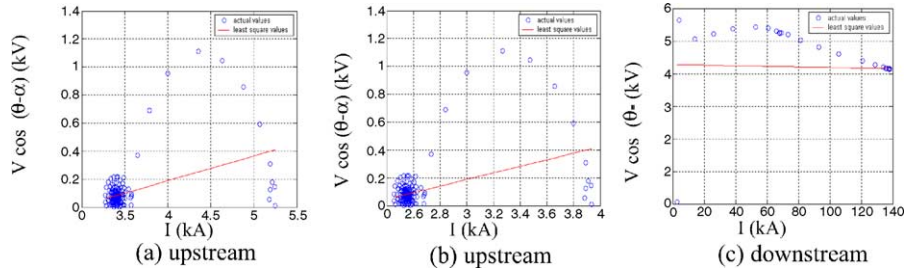


Fig. 12. Fault at F2 (SLG) for monitoring at (a) M1, (b) M2 and (c) M3 using the slope method.

tive values at the beginning of the sag, indicating the correct upstream locations of voltage sag source. Whereas in Fig. 11(c), the real current settles out to a positive value at the beginning of the sag, indicating a downstream voltage sag source. Fig. 12(a)–(c) shows the results from the method of system trajectory slope [5]. Both results in Figs. 12(a) and (b) show a positive slope which indicates an upstream location of voltage sag source. In Fig. 12(c) a negative slope is observed to indicate a downstream location of voltage sag source.

A DLG fault is simulated for phases A and B at F2. Fig. 13(a)–(c) shows the results of phase A current measurements. Fig. 13(a) and (b) shows that the real current settles out to negative values at the beginning of the sag, indicating the correct location of voltage sag source which is from upstream. Fig. 13(c) shows that the real current settles out to positive values at the beginning of the sag, indicating the voltage sag source is from downstream. Fig. 14(a)–(c) shows the results from the method of system trajectory slope [5]. In Fig. 14(a) and (b) positive slopes

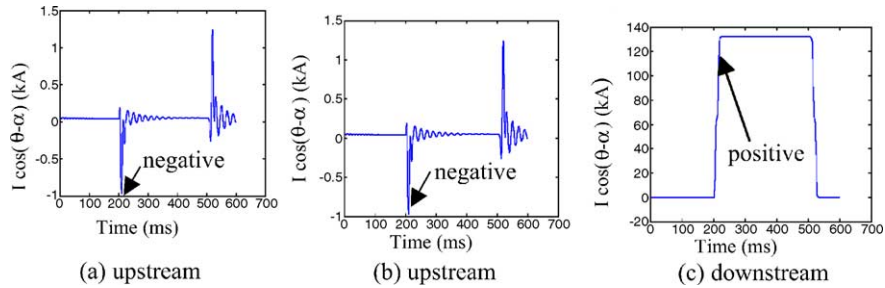


Fig. 13. Fault at F2 (DLG, phase A) for monitoring at (a) M1, (b) M2 and (c) M3 using the real current component.

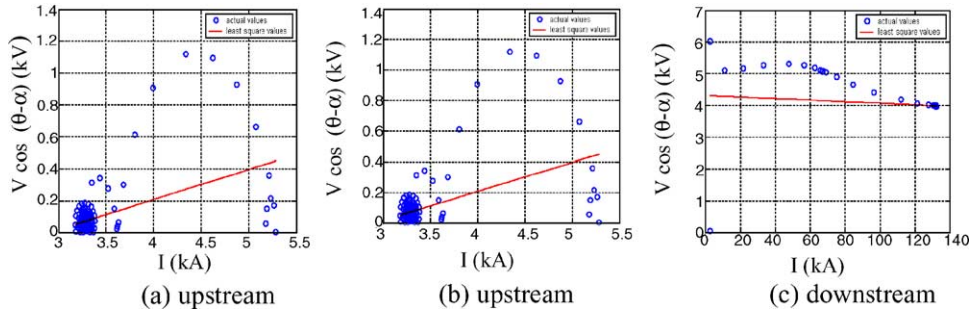


Fig. 14. Fault at F2 (DLG, phase A) for monitoring at (a) M1, (b) M2 and (c) M3 using the slope method.

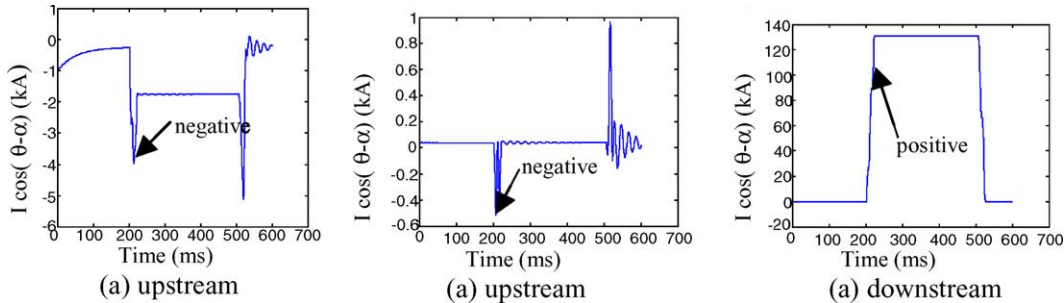


Fig. 15. Fault at F2 (DLG, phase B) for monitoring at (a) M1, (b) M2 and (c) M3 using the real current component.

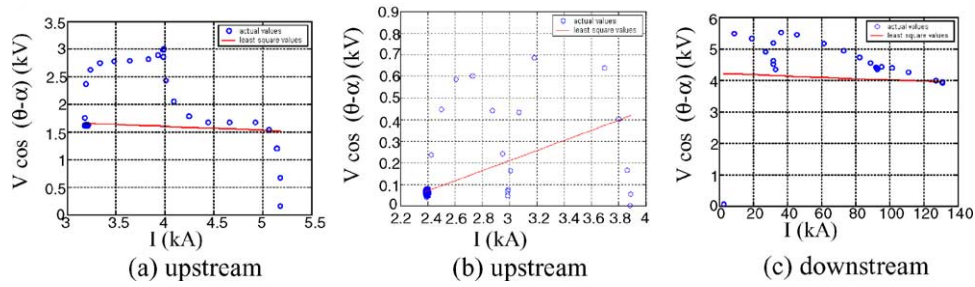


Fig. 16. Fault at F2 (DLG, phase B) for monitoring at (a) M1, (b) M2 and (c) M3 using the slope method.

are noticed, thus indicating upstream locations of voltage sag source. Meanwhile in Fig. 14(c) a negative slope is observed, indicating a downstream location of voltage sag source.

Results of phase B current measurements for a DLG fault at F2 are presented in Fig. 15(a)–(c) and Fig. 16(a)–(c) for the proposed method and the system trajectory method, respectively. In Fig. 15(a) and (b) the real current settles out to a negative value at the beginning of the sag which indicates that the voltage sag source is from upstream. In Fig. 15(c), the real current settles out to a positive value at the beginning of the sag, indicating the voltage sag source is from downstream. In Fig. 16(a), a negative slope is noticed indicating the voltage sag source is from downstream location. This result contradicts with the actual event in which the voltage sag source is actually from upstream. However Fig. 16(b) shows a positive and Fig. 16(c) shows a negative slope, indicating upstream and downstream voltage sag source, respectively, and the results agree with the proposed real current component method.

Table 3

Balanced fault bus details for two-source system

Fault bus	Monitoring buses	
	Downstream fault	Upstream fault
FA1	MA1	MA2, MB1, MB2
FB1	MB1	MB2, MA1, MA2

4.2. Results from two-source system

For a two-source system, a balanced fault is simulated at FA1 and FB1. The results from two-source system of the proposed method and the slope method are first presented and analyzed.

4.2.1. Case A: balanced fault

Table 3 tabulates the faulted bus locations.

Fig. 17(a)–(c) shows the graph of real current components plotted against time for a fault at FA1 measurements made at MA1, MA2 and MB1, respectively. In Fig. 17(a), it can be seen that the real current settles out to a positive value at

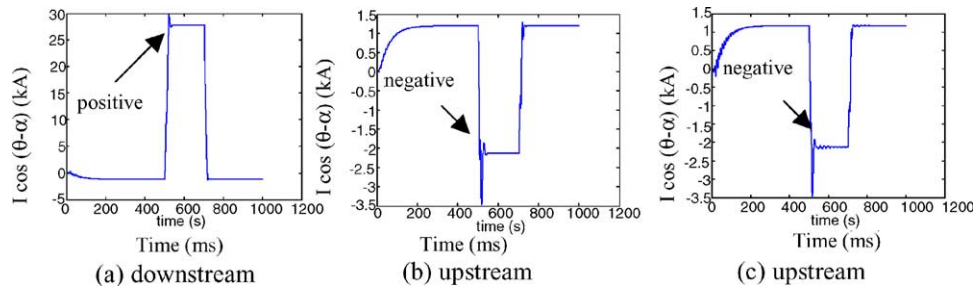


Fig. 17. Balanced fault at FA1 for monitoring at (a) MA1, (b) MA2 and (c) MB1 using the real current component.

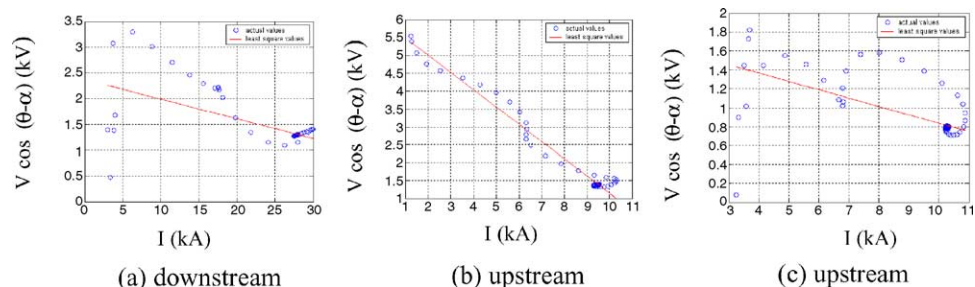


Fig. 18. Balanced fault at FA1 for monitoring at (a) MA1, (b) MA2 and (c) MB1 using the slope method.

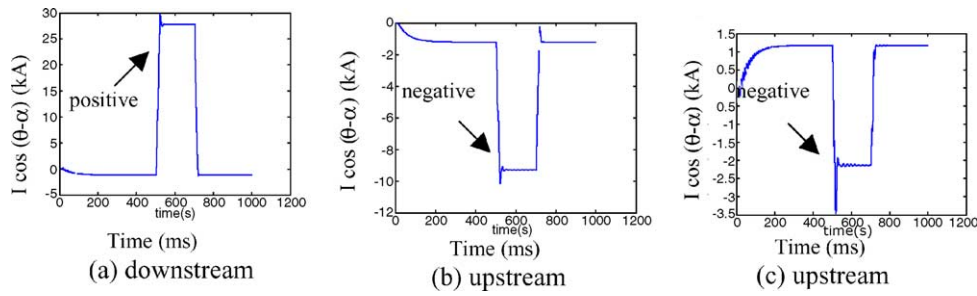


Fig. 19. Balanced fault at FA2 for monitoring at (a) MB1, (b) MB2 and (c) MA1 using the real current component.

the beginning of the sag, indicating the voltage sag source is from downstream. Fig. 17(b) and (c) shows that the real current settles out to a negative value at the beginning of sag, indicating that the voltage sag source is from upstream. Fig. 18(a)–(c) shows the results from the method of system trajectory slope [5]. In Fig. 18(a)–(c) a negative slope is noticed indicating the voltage sag source is from downstream location. Results from Fig. 18(b) and (c) contradicts with the actual event in which the voltage sag source is actually from upstream.

Fig. 19(a)–(c) shows the graph of real current components plotted against time for a fault at FA2 measurements made at MB1, MB2 and MA1, respectively. In Fig. 19(a), it can be seen that the real current settles out to a positive value from the beginning of the sag, indicating the voltage sag source is from downstream. Fig. 19(b) and (c) shows that the real current settles out to a negative value at the beginning of sag, indicating that the voltage sag source is from upstream. Fig. 20 shows the results from the method of system trajectory slope [5]. In Fig. 20(a)–(c), a negative slope is noticed

Table 4

Unbalanced fault bus details for two-source system

Fault bus	Monitoring buses	
	Downstream fault	Upstream fault
FA1	MA1	MA2, MB1, MB2
FA2	MB1	MB2, MA1, MA2

indicating the voltage sag source is from a downstream location. Hence results from Fig. 20(b) and (c) contradict with the actual event in which the voltage sag source is actually from upstream.

4.2.2. Case B: unbalanced fault

The results of unbalanced faults created at FA1 and FA2 are first presented and analyzed. Table 4 tabulates the faulted bus locations.

Fig. 21(a)–(c) shows the graph of real current components plotted against time for a fault at FA1 measurements made at MB1, MB2 and MA1, respectively. In Fig. 21(a),

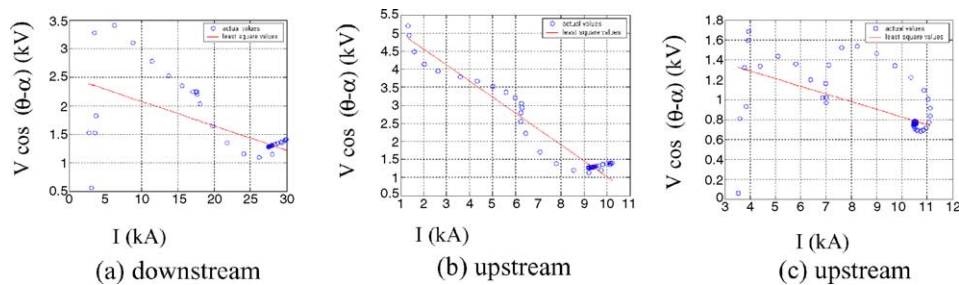


Fig. 20. Balanced fault at FA2 for monitoring at (a) MB1, (b) MB2 and (c) MA1 using the slope method.

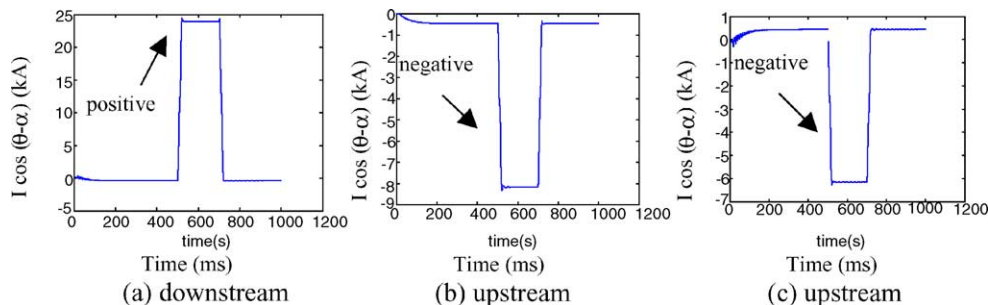


Fig. 21. Unbalanced fault at FA1 for monitoring at (a) MB1, (b) MB2 and (c) MA1 using the real current component.

Table 5

Comparison of voltage sag location methods using DP/DE, SM and RCM

Method	DP/DE	SM	RCM
1	Requires calculation of power and energy for the entire sag duration	Requires calculation of $V \cos(\theta - \alpha)$ and I for the entire sag duration	Requires only $I \cos(\theta - \alpha)$ at the beginning of sag
2	Requires calculation of the difference between instantaneous power before, during and after the sag	Require line-fitting method to plot the gradient in which the results based on the accuracy of this line-fitting is used to determine the sag source location	Does not require line-fitting method or calculation of difference between power and energy which means less computational time is needed to determine the sag source location
3	Only proved applicable in one-source system	Only proved applicable in one-source system	Applicable in one-source and two-source system

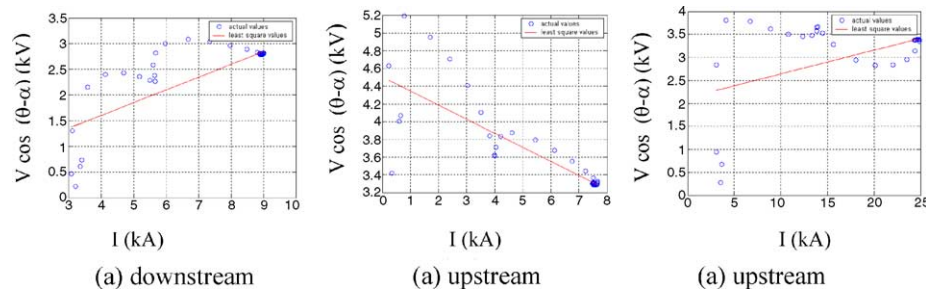


Fig. 22. Unbalanced fault at FA2 for monitoring at (a) MB1, (b) MB2 and (c) MA1 using the slope method.

it can be seen that the real current settles out to a positive value from the beginning of the sag, indicating the voltage sag source is from downstream. Fig. 21(b) and (c) shows that the real current settles out to a negative value at the beginning of sag, indicating that the voltage sag source is from upstream. Fig. 22 shows the results from the method of system trajectory slope [5]. In Fig. 22(a), a positive negative slope is noticed indicating the voltage sag source is from upstream location. This result does not correctly indicate the voltage sag location in which it is from downstream. Fig. 22(b) also does not locate the voltage sag source because it has negative slope. For this unbalanced fault, only result in Fig. 22(c) can locate the voltage sag source correctly from upstream location by its positive slope.

4.3. Comparison of the disturbing power/disturbing energy (DP/DE), slope method (SM) and the real current component method (RCM)

For the real current component method, the real current is plotted against time in which the beginning of the sag can be seen to occur at 0.2 s in a one-source system and at 0.5 s in a two-source system. Whereas using the slope method, the voltage component is plotted against current component in which no time scale is involved. Thus, the consistency of the algorithm with respect to time is only available in the real current component graphs. Table 5 presents the summary of the advantages and disadvantages of the disturbing power/disturbing energy method, slope of system trajectory method and the real current component method. The real

current component has advantages over the other two methods because it requires less parameter for index calculation, less computational time for data processing and it also has been proven to work well in both balanced and unbalanced system for one-source and two-source system by simulations in this paper.

5. Conclusion

This paper has presented a new method to detect the voltage sag source as seen at the monitoring point by examining the direction of the real current component at the beginning of a sag. From the results, it has been proven that the proposed real current component method satisfy 100% of the voltage sag source location for the three-phase balanced and unbalanced faults for the one-source system. The slope of system trajectory method however satisfies only 89% of the three-phase balanced fault and unbalanced fault as discussed in the paper for the one-source system. The method has also been verified for the two-source system in which it has given very promising results. The new method has proven to be more accurate compared with the slope of system trajectory method. The characteristics of the proposed method can be listed as follows:

- It only requires three parameters for calculations, namely the magnitude of current and the phase angles of voltage and current at the monitoring point.
- It does not require a line-fitting method as in the slope of system trajectory method, hence straightforward and less

computational time is required in the computation of the real current component indicator.

- It is proven to work well with not only balanced three-phase fault but also with unbalanced faults (SLG and DLG) in a one-source system.
- It is proven to work well with not only balanced three-phase fault but also with unbalanced faults for two-source system.

Acknowledgements

The authors gratefully acknowledge the Universiti Teknologi MARA for financial support in terms of scholarship and Universiti Kebangsaan Malaysia (IRPA code 02-02-02-0014-EA188) for financial support on the project.

References

- [1] M.H.J. Bollen, Voltage sags in three-phase systems, *IEEE Power Eng. Rev.* 17 (2001) 8–15.
- [2] M.H.J. Bollen, *Understanding Power Quality Problems*, IEEE Press, 2000, pp. 1–34.
- [3] M.F. McGranaghan, D.R. Mueller, Voltage sags in industrial systems, *IEEE Trans. Ind. Appl.* 29 (2) (1993) 397–403.
- [4] IEEE Std. 1159-1995: Recommended Practice for Monitoring Electric Power Quality. ISBN 1-55937-549-3.
- [5] C. Li, T. Tayjasanant, W. Xu, X. Li, Method for voltage sag source detection by investigating slope of the system trajectory, *IEE Proc. Gen. Transm. Distrib.* 150 (3) (2003) 367–372.
- [6] A.C. Parsons, W.M. Grady, E.J. Powers, J.C. Soward, A Direction finder for power quality disturbances based upon disturbance power and energy, *IEEE Trans. Power Delivery* 15 (3) (2000) 1081–1086.
- [7] W.D. Stevenson, *Elements of Power System Analysis*, 4th ed., McGraw-Hill, 1982.
- [8] The Electricity Council, *Power System Protection 1*, Peter Peregrinus Ltd., UK, 1986.

Biographies

N. Hamzah (M'2001–2002, StudentM2003) received her B.Eng.(Hon) and M.Sc. (power system) from University of Wales Institute of Science and Technology in 1988 and University of Malaya, Malaysia in 1993, respectively. She is currently enrolled as a Ph.D. student at Universiti Kebangsaan Malaysia, Malaysia.

A. Mohamed (Senior Member 2003) received her B.Sc.Eng. from King's College, University of London in 1978 and M.Sc. and Ph.D. (power system), from University of Malaya, Malaysia in 1988 and 1995, respectively. She is currently a professor at Universiti Kebangsaan Malaysia (UKM), Malaysia. Her current research interests are in power quality and other power system studies.

A. Hussain (Member 1997) received her B.Sc. in electrical engineering, Louisiana State University, USA, M.Sc. in systems and control, UMIST, UK, and Ph.D. from Universiti Kebangsaan Malaysia (UKM) in 1985, 1989 and 1997, respectively. She is currently an Associate Professor at UKM. Her research interests are signal processing, neural networks and their applications, which include power quality studies. She is also a member of Tau Beta Pi.